

ANKARA MURTED, Turkey

WMO: 171270

Lat: 40.079N

Lon: 32.566E

Elev: 2767

StdP: 13.28

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.3	14.4	6.4	8.5	14.8	10.0	10.2	19.1	19.3	34.2	17.1	34.0	2.7	0	0.521

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	28.5	95.0	62.9	91.8	62.5	89.4	62.2	66.6	85.0	65.3	83.9	64.1	82.8	8.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.9	88.4	72.5	59.1	83.0	71.6	57.5	78.1	70.2	32.8	84.7	31.7	83.7	30.7	83.0	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4) 19.6	16.6	14.3		4.6	99.5	4.9	2.7	1.0	101.5	-1.8	103.1	-4.6	104.6	-8.1	106.6	
(5)			DB	2.1	69.0	6.0	2.4	-2.2	70.8	-5.7	72.2	-9.1	73.6	-13.4	75.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.9	32.1	36.1	42.1	50.1	59.4	67.1	73.9	74.8	65.3	54.7	42.2	35.3	(6)
	DBStd	16.00	8.34	7.80	6.07	5.95	5.62	5.05	4.77	4.37	6.39	6.47	6.30	7.53	(7)
	HDD50	1997	553	390	252	69	2	0	0	0	0	31	243	457	(8)
	HDD65	5209	1018	810	710	446	189	34	2	0	74	321	684	921	(9)
	CDD50	3047	0	1	7	73	294	514	742	770	458	177	10	1	(10)
	CDD65	782	0	0	0	0	16	98	279	305	82	2	0	0	(11)
	CDH74	10486	0	0	0	28	418	1395	3537	3702	1274	132	0	0	(12)
	CDH80	4539	0	0	0	0	99	491	1677	1786	473	13	0	0	(13)
(14) Wind	WSAvg	5.8	4.4	5.1	6.3	6.2	6.0	6.5	7.8	7.5	6.3	4.9	4.0	4.2	(14)
(15) Precipitation	PrecAvg	14.8	1.7	1.3	1.3	1.7	1.9	1.3	0.6	0.5	0.7	0.9	1.1	1.9	(15)
	PrecMax	24.6	3.9	3.5	3.3	3.7	3.5	3.6	2.0	1.7	3.7	2.7	4.6	3.8	(16)
	PrecMin	9.8	0.2	0.3	0.3	0.2	0.7	0.1	0.0	0.0	0.0	0.0	0.1	0.2	(17)
	PrecStd	3.1	1.1	0.9	0.8	1.0	0.7	1.1	0.6	0.4	0.8	0.7	0.9	0.9	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.7	63.4	68.3	78.4	86.3	93.2	100.0	100.0	93.5	81.0	69.4	57.3	(19)
		MCWB	44.7	48.0	49.7	55.4	59.7	62.3	63.4	64.1	60.2	56.6	51.7	49.3	(20)
	2%	DB	49.8	57.3	64.0	73.0	82.6	89.3	95.0	95.3	89.5	77.4	64.3	52.1	(21)
		MCWB	43.0	45.6	47.8	53.3	59.1	62.2	63.3	63.1	60.0	55.8	50.3	46.3	(22)
	5%	DB	46.4	53.3	60.4	68.9	78.8	85.9	91.7	92.8	85.8	73.8	59.2	49.6	(23)
		MCWB	41.5	43.8	46.3	51.8	58.1	61.8	63.0	63.0	59.1	55.0	48.3	45.1	(24)
	10%	DB	43.0	48.4	55.8	64.8	74.9	82.3	89.2	89.4	81.9	69.7	55.2	46.4	(25)
		MCWB	40.1	41.7	45.1	50.1	57.0	60.8	62.7	62.3	58.1	53.9	46.1	43.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.1	49.9	51.6	56.7	63.1	66.4	68.4	68.9	64.3	59.3	54.8	50.7	(27)
		MCDB	50.8	60.4	62.3	71.3	78.9	83.2	86.7	88.0	81.8	70.7	64.9	55.3	(28)
	2%	WB	43.8	46.8	49.1	54.9	61.0	64.8	66.6	67.1	62.5	58.1	51.9	47.5	(29)
		MCDB	47.9	54.6	60.3	69.1	76.3	81.3	85.2	85.6	80.9	70.9	59.8	51.2	(30)
	5%	WB	42.0	44.6	47.4	53.1	59.7	63.6	65.3	65.5	61.0	56.7	49.3	45.5	(31)
		MCDB	45.2	51.7	57.5	65.8	74.5	79.8	84.4	84.4	79.3	70.1	57.7	48.6	(32)
	10%	WB	40.1	42.4	45.7	51.5	58.1	62.3	64.0	64.2	59.7	55.6	47.3	43.6	(33)
		MCDB	42.9	48.1	54.6	63.0	72.1	78.0	83.4	83.3	77.3	67.5	54.1	46.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	15.1	20.3	23.1	25.3	26.2	26.6	28.6	28.5	28.7	26.6	24.4	16.4	(35)
		MCDBR	21.1	28.4	32.3	32.7	32.9	32.0	31.4	32.2	34.2	33.6	31.5	20.3	(36)
	5% WB	MCWBR	14.7	17.6	18.3	16.4	13.9	11.2	9.8	10.4	12.8	15.9	19.0	14.4	(37)
		MCWBR	17.7	24.3	27.0	28.4	29.1	28.0	28.4	27.8	28.4	27.4	26.7	17.7	(38)
(40) Clear-Sky Solar Irradiance	taub		0.303	0.328	0.371	0.407	0.387	0.368	0.365	0.363	0.345	0.344	0.314	0.304	(40)
		taud	2.382	2.332	2.238	2.168	2.291	2.363	2.369	2.353	2.402	2.403	2.456	2.439	(41)
	Ebn at Noon		272	280	278	275	283	287	287	285	284	271	267	263	(42)
		Edh at Noon	27	33	40	46	42	39	38	37	33	30	25	23	(43)
(44) All-Sky Solar Radiation	RadAvg		594	908	1269	1625	1932	2195	2335	2078	1633	1114	772	528	(44)
	RadStd		73	133	115	148	145	194	103	94	115	81	78	68	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page